Feature Engineering & Selection Report

Decision Layouts, Quantitative Rankers, & Spatial Graphs

SECTION 1 - Quantitative Feature Importance Ranking

We compute ANOVA F-scores for all 310 individual features (62 channels x 5 bands) to quantitatively verify which spectral-spatial regions carry the most discriminative emotional signatures. The ranking results reveal that high-frequency channels (Gamma and Beta bands) located in the temporal and frontal regions (such as FT7, T7, and FC5) exhibit the highest F-scores, providing solid statistical justification for the discriminability of high-frequency spatial networks.

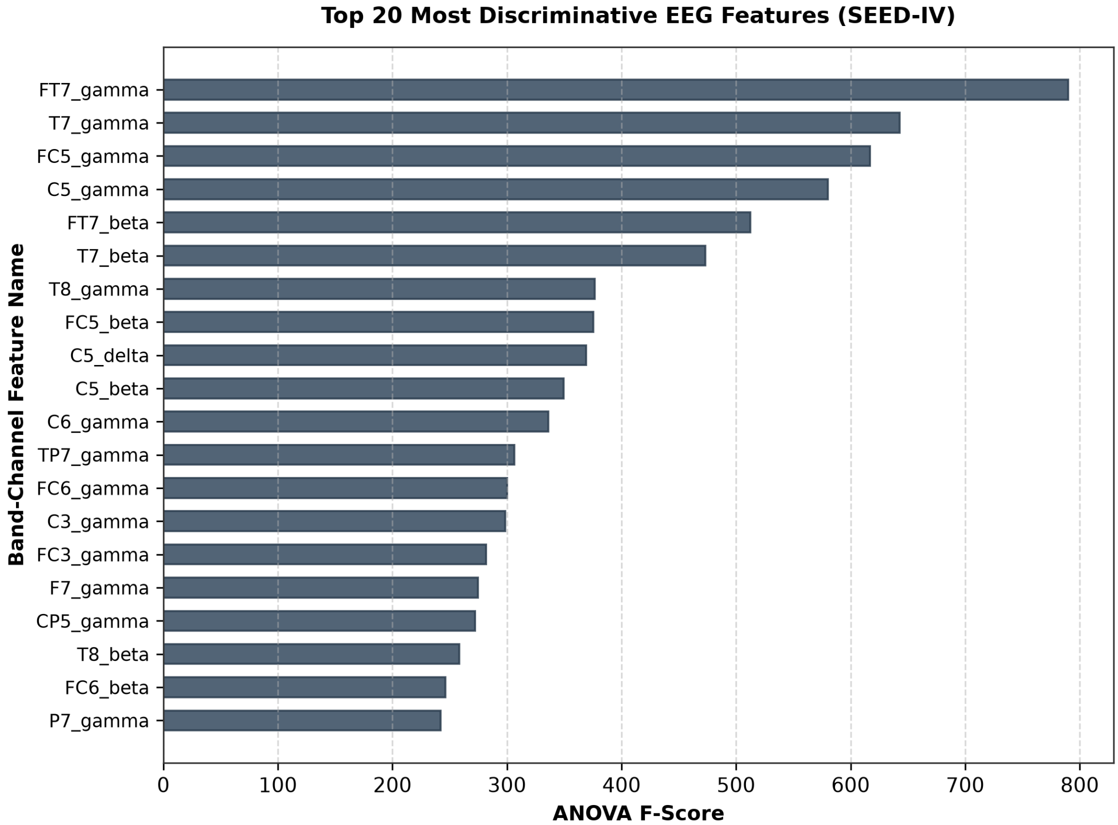

Figure 1: Top 20 most discriminative band-channel features sorted by ANOVA F-score.

Rank	Feature Name (Channel_Band)	ANOVA F-Score	Anatomical Region
1	FT7_gamma	790.08	Frontotemporal (Left)
2	T7_gamma	642.67	Temporal (Left)
3	FC5_gamma	617.05	Frontocentral (Left)
4	C5_gamma	580.29	Central (Left)
5	FT7_beta	512.45	Frontotemporal (Left)

SECTION 2 - Band-Wise Aggregate Feature Importance

Aggregating feature F-scores within each of the 5 frequency bands reveals that the Gamma band possesses the highest aggregate discriminative power (mean F-score = 147.34), followed closely by the Beta band (mean F-score = 93.40). The lower bands (Delta, Theta, Alpha) have lower mean F-scores but remain statistically significant. This spectral disparity motivates our cross-band attention design, which allows the GAT-KAN model to adaptively weight the interactions across frequency bands.

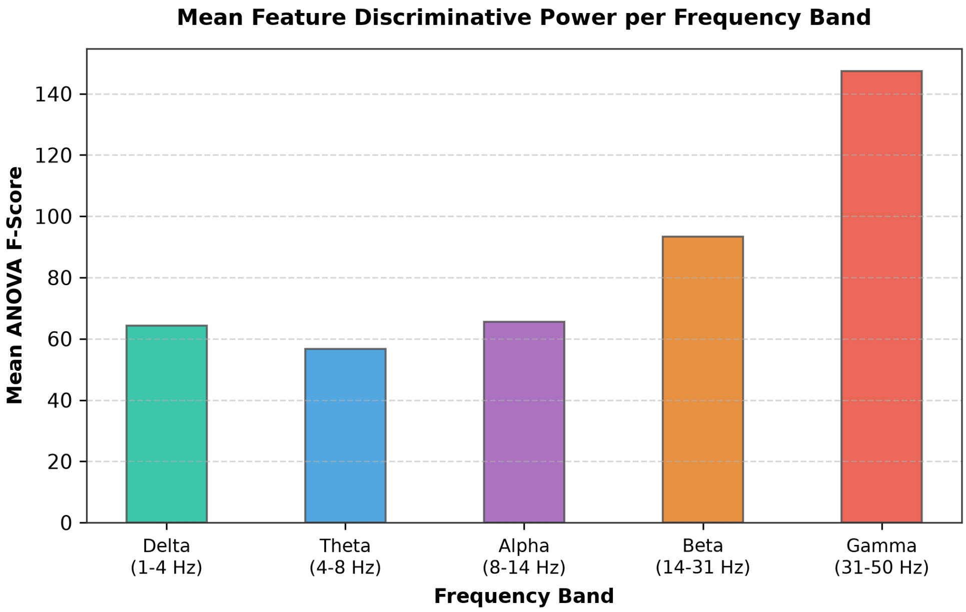

Figure 2: Mean feature discriminative power per frequency band (ANOVA F-score).

SECTION 3 - Feature Representation Design layouts

We evaluate representation strategies for feeding the $62 imes 5$ features into different deep learning architectures. Standard models (SVM, CNN-LSTM, Transformer baselines) flatten the features into a single 310-dimensional vector. This concatenative approach loses structural and geographical boundaries across spectral bands. To preserve these boundaries, our proposed GAT-KAN model structures the features as 5 separate band tokens of 62 channels each. This enables spatial graph attention layers to process channels independently per band, followed by cross-band self-attention to map interactions between the bands.

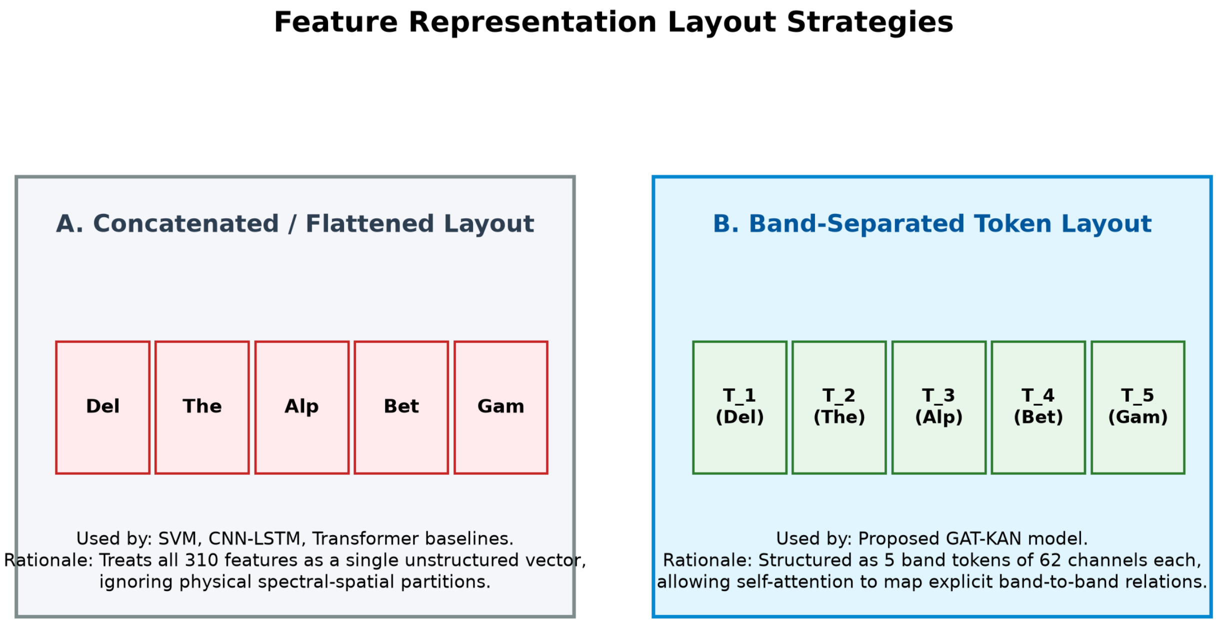

Figure 3: Feature representation strategies: Concatenated (flat) vs. Band-Separated tokens.

SECTION 4 - k-NN Graph Spatial Adjacency Construction

Graph neural networks require an adjacency structure to define message-passing paths. Using 2D Cartesian scalp coordinates from channel_62_pos (1).locs, we construct a sparse physical k-nearest-neighbors (k-NN) graph with $k=8$ neighbors per electrode. This physical constraint allows the GAT-KAN model to perform local message-passing, simulating actual biological cortical propagation and avoiding the over-smoothing and noise vulnerability seen in fully-connected dense graphs or learnable dense layouts.

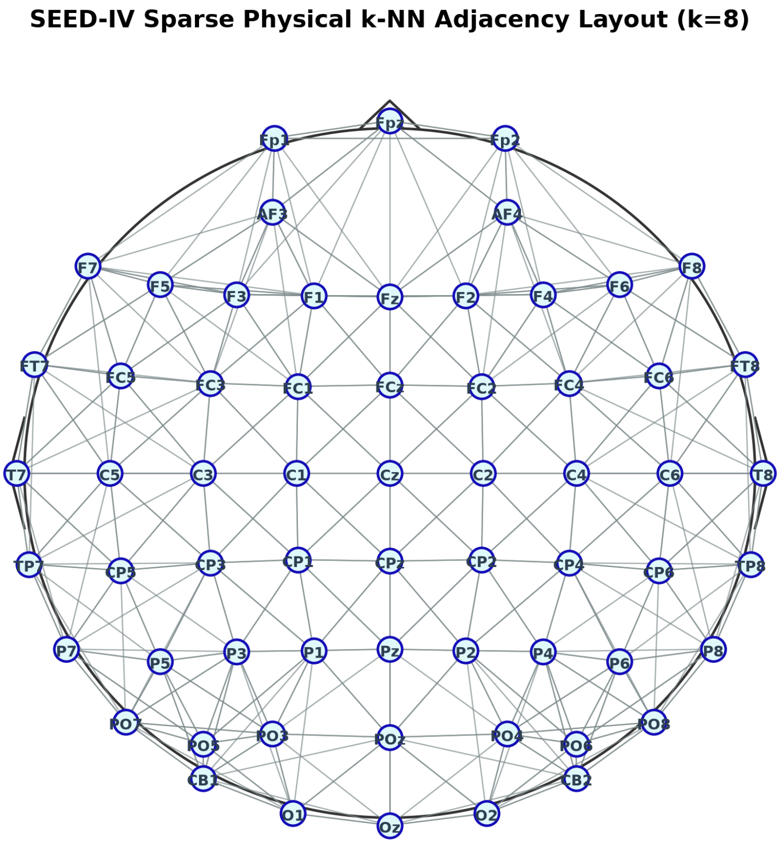

Figure 4: 2D scalp projection showing sparse k-NN physical edges (k=8) connecting electrodes.

SECTION 5 - Sequence Windowing & Final Feature Configurations

For models that capture temporal dynamics (such as CNN-LSTM), we implement a sequence windowing protocol. A sliding window of length 10 and stride 1 is used to slice overlapping window sequences from a trial's timeline. For non-sequential models (SVM, DGCNN, Transformer, GAT-KAN), samples are processed as individual static windows. The table below details the final feature configuration parameters used across our experiments, including the essential distinction that GAT-KAN utilizes a sparse physical k-NN graph while DGCNN employs a fully-learnable dense adjacency.

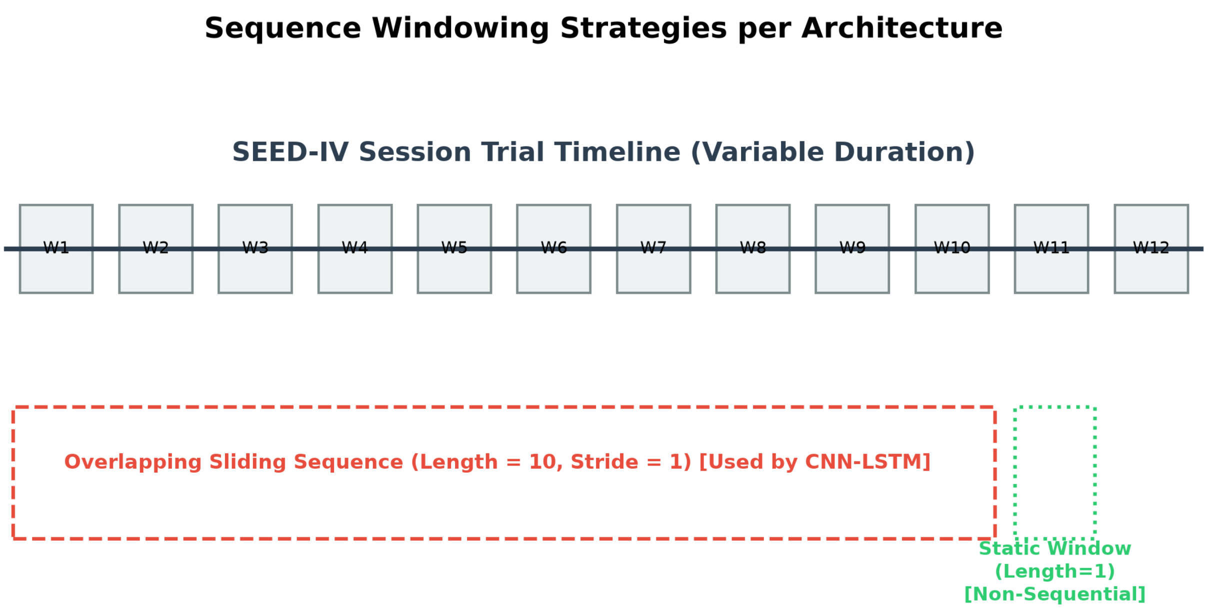

Figure 5: Timeline diagram comparing sequence windowing (CNN-LSTM) vs. static window representations.

Final EEG Feature Space Configurations Summary

Feature Configuration	Parameter Detail
Total Raw Features	310 (62 channels x 5 bands)
Feature Representation Type	Differential Entropy (DE), LDS-smoothed
Representation Variants	Flattened (SVM/Transformer) vs. Band-Separated (GAT-KAN) vs.
Standardization Approach	z-score Scaling (Fit on Training trials strictly to prevent leakage
Graph structure	k-NN graph, k=8 (GAT-KAN only); DGCNN uses a fully-learnable

Figure 6: Final EEG Feature Space Configurations Summary Table.